



Vectors · Worksheet 1

Core Pure 1 Chapter 9: Vectors. Answer all questions. Show your working clearly. Total: 85 marks. Time: 105 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 9. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- λ , μ and t are scalar parameters. Give exact answers where possible; give angles in degrees to 1 decimal place and other non-exact answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Lines and the scalar product	20	
B Equations of planes and the vector product	20	
C Intersections and angles with planes	20	
D Distances: point to plane and skew lines	20	
Roar-level challenge	5	
Total	85	

Section A Lines and the scalar product

20 marks

1. Find a vector equation of the line through the points $A(1, -2, 3)$ and $B(4, 0, 1)$. [2]

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2. Find the acute angle between the lines with direction vectors $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 4 \\ -1 \end{pmatrix}$. [3]

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3. The line l passes through the point $(2, -1, 4)$ and has direction vector $\begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix}$. [3]

Find a Cartesian equation of l , and determine whether the point $(8, 9, 0)$ lies on l .

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4. Show that the lines [4]

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad \text{and} \quad l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

intersect, and find the coordinates of their point of intersection.

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5. The lines l_1 and l_2 have equations

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad \text{and} \quad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

(a) Show that l_1 and l_2 do not intersect.

[4]

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(b) Explain why l_1 and l_2 are skew.

[1]

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(c) Find the acute angle between l_1 and l_2 .

[3]

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(Total for Question 5 is 8 marks)

**Section B Equations of planes and the vector product**

20 marks

6. Find a Cartesian equation of the plane that passes through the point $(2, -1, 3)$ and has normal vector [2]

$$\begin{pmatrix} 4 \\ 1 \\ -2 \end{pmatrix}.$$

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7. Find $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$, and verify that your answer is perpendicular to both vectors. [3]

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8. The plane Π has vector equation $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$. [3]

Using your answer to Question 7, find an equation of Π in the form $\mathbf{r} \cdot \mathbf{n} = p$, and hence a Cartesian equation of Π .

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9. Find a Cartesian equation of the plane that passes through the points $A(1, 0, 1)$, $B(2, 3, 0)$ and $C(0, 1, 4)$. [4]

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10. The line l has equation $\mathbf{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and the point P has coordinates $(3, 2, 4)$. The plane Π contains l and P .

(a) Find a vector perpendicular to both $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}$. [2]

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(b) Hence find a Cartesian equation of Π . [3]

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(c) Write down a vector equation of Π in the form $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b} + \mu \mathbf{c}$. [1]

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(d) Show that the line m with equation $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ -7 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ lies in Π . [2]

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(Total for Question 10 is 8 marks)



Section C Intersections and angles with planes

20 marks

11. Show that the line $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is parallel to the plane $x - 2y + z = 7$ but does not lie in it. [2]

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12. Find the coordinates of the point where the line $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ meets the plane $x + 2y + z = 11$. [3]

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13. Find the acute angle between the line $\mathbf{r} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ and the plane $2x - y + 2z = 5$. [3]

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14. The plane Π has equation $x + 2y + 2z = 1$ and the point P has coordinates $(5, 3, 4)$. [4]
Find the coordinates of the foot of the perpendicular from P to Π , and hence the coordinates of the reflection of P in Π .

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15. The points A and B have coordinates $(1, 2, 3)$ and $(3, 3, 1)$. The line l passes through A and B . The plane Π has equation $x + 4y + 8z = 13$.

(a) Find a vector equation of l .

[2]

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(b) Find the coordinates of the point where l meets Π .

[3]

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(c) Find the acute angle between l and Π .

[3]

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(Total for Question 15 is 8 marks)

**Section D Distances: point to plane and skew lines**

20 marks

16. Find the shortest distance from the point $(1, 2, 3)$ to the plane $2x - y + 2z = 10$. [2]

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17. Find the distance between the parallel planes $2x + 3y - 6z = 4$ and $2x + 3y - 6z = -10$. [3]

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18. The point $(k, 1, 2)$ is at a distance of 3 units from the plane $2x + y - 2z = 1$. [3]
Find the possible values of k .

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19. Find the shortest distance between the lines [4]

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

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20. In a model of a building site, a water pipe runs along the line l_1 and an electric cable runs along the line l_2 , where

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad l_2: \mathbf{r} = \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}.$$

The units are metres.

- (a) Find a vector that is perpendicular to both l_1 and l_2 . [2]

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- (b) Find the shortest distance between the pipe and the cable. [3]

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- (c) The plane Π contains l_1 and is parallel to l_2 . Find a Cartesian equation of Π . [2]

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- (d) State one limitation of the model. [1]

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(Total for Question 20 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. The line l has equation $\mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ and the plane Π has equation $2x - y + 2z = 1$. [5]

Show that l is not parallel to Π , and find the coordinates of the two points on l that are at a distance of 3 units from Π .

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END OF WORKSHEET

Total: 85 marks